Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 59 (2015) 19 – 26

Procedia Computer Science

International Conference on Computer Science and Computational Intelligence (ICCSCI 2015)

A Novel Broadband Line Construction Using 3G Service

Lukas Tanutama[a], Rico Wijaya[a], Kevin Alexander[a], Agung Wiratama[a], Raphael Indraswara[a]

[a]*Bina Nusantara University, Computer Engineering, Jl.KH. Syahdan 9, Jakarta 11480, Indonesia*

Abstract

Quite often, it is necessary to upload or download large files in a relatively short time and from or to locations that do not necessarily have adequate infrastructure. If the location is static the needs can be addressed easily. There are high-speed transmission or broadband services offered by a large number of operators through both conventional and fiber optic cables. If the users are moving nomadically or are mobile, then obtaining large-capacity transmission systems is not simple. For some purposes, especially for small and medium-sized companies, they need to have alternative solutions that are economic and technically feasible. Cellular service providers offer 3G Internet services with wide coverage. This study develops an application using 3G services to provide broadband access that is economic and relatively simple. This solution uses multiple 3G connection to transmit information at high speed as if there is broadband channel capacity. Information on a Web Server can be downloaded by the parties concerned. The information retrieval can obtain the necessary capacity using multiple 3G connections. The technique of bundling multiple lower speed channels to obtain high-speed channel is called inverse multiplexing. In this research, the technique is simplified by using a router that has the ability to provide load balancing. This research was carried out by conducting laboratory experiments and field measurements. The experimental results show that the concept of inverse multiplexing through a number of channels with 3G services can run well for downloading files.

© 2015 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).
Peer-review under responsibility of organizing committee of the International Conference on Computer Science and Computational Intelligence (ICCSCI 2015)

Keywords: Inverse Multiplexing, NHT, Mangler, 3G

* Corresponding author. Tel.: +62-21-5345830
E-mail address:lukast12@binus.edu

1877-0509 © 2015 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).
Peer-review under responsibility of organizing committee of the International Conference on Computer Science and Computational Intelligence (ICCSCI 2015)
doi:10.1016/j.procs.2015.07.333

1. Introduction

Broadband transmission is a must for information and communication technology considering the amount of information that must be transmitted. The file size that is currently being exchanged continues to expand due to the type of information it contains. Current information is characteristically multimedia and not only text. The electronic devices are now able to handle multimedia information and fulfilling the expected quality. To transmit this kind of information a transmission system that has adequate speed or capacity is necessary to avoid unnecessary accumulation of data. Inadequate speed will delay the transmission of large amounts of data that are generated by the source. Multimedia information will be disrupted even cannot be used due to the delay or loss of data that should arrive at the receiver. To reduce the possibility of disruption of the information due to the long time delay or loss of data during transmission, various algorithms and standard provisions have been developed to minimize the amount of data without reducing quality. The efforts still generates relatively large files; hence, the transmission of large files cannot be avoided. To transmit these type of information the ability to deliver data with high-speed is a must. The transmission must meet a number of stringent technical requirements in order to keep the high quality at receiver. For real time data, it is likely that the disturbed information particularly resulted in the delay of processing the information received at the receiver. To overcome this problem, the capacity of the transmission line should be increased. However, the capacity of a commercial transmission medium is often also inadequate. It needs quite a long time to send the information file from a source to its destination. If the transmission medium is copper, the obstacles are not only due to the limited capacity but also due to the distance, geographical location, and the availability of these facilities both at the source and receiver locations. Sometimes a number of sites are connected with optical fiber infra structure that enable high-speed transmission. High-speed transmission with optical fiber is very limited. Hence, it often fails to meet the demands for high-speed transmission. The main limitation is the very limited availability. Geographical constraint is also one of the reasons for limited availability. In certain circumstances the geographical constraint can be overcome by utilizing satellite communications services. Television broadcasts often use satellite transmission system for real time news programs. For factual and reliable information, it is only logical that such information is in the form of pictures, video or live television coverage. For this purpose almost every television station has Satellite News Gathering (SNG). SNG car serves as a sender to the Master Control Room (MCR) through a satellite connection. The problem o SNG is the investment financial outlay required and the huge operational expenses. The costs do not include cost of transmitting the program. In major cities broadband transmission line service can be obtained quite easily. In certain locations the transmission system availability constraints make it difficult to disseminate high volume information as well as its acquisition, especially if the information value time limited. Retrieval and dissemination of multimedia information particularly broadband files within a short time will be hampered if broadband infrastructure is not available. The need for broadband will be felt if the transmitted information is real time video. Moreover, there are various other uses that require high-speed transmission, but no facilities that enable it. Broadband transmission requires the availability of adequate physical infrastructure such as fiber optic, satellite or microwave. Satellite and microwave require high operating costs in line with the capacity needed. Transmission via cable is based on ADSL, cable modems or fiber optics. Cable transmissions are for a fixed location and usually are not readily available at certain locations. Microwave transmission is limited by the availability of the frequency spectrum and license to use the frequencies.

Cellular telephone communication techniques have grown rapidly and have nationwide coverage. The latest generation of mobile communication emphasizes on broadband Internet data communications with nationwide coverage. The third generation of mobile communications is known as 3G communication system. 3G data communications has the flexibility as cellular telephone communications in general. 3G communication connects users to the Internet. 3G is the latest technology which can connect devices to the Internet economically and have reasonably high-speed .This research developed a system to obtain high-speed broadband services through the use of a number of 3G channels. Technically, the system is interesting due to the fact that the information is transmitted in digital form. A number of 3G channel is used to transmit data package from the Web server on the Internet to get the desired information from the Web server. A router will be configured such that broadband capability can be obtained. The system will have a bandwidth equal to the number of 3G channels used. 3G channel capacity will depend on the type of service subscribed from the operator.

Many studies examining the success in the use of 3G network services, numerous other studies argue that there will be no contender or a substitute for 3G services in the near future. 3G as today's technology can easily connect devices to the Internet and is accessible to everyone. The above facts underlie the role of 3G in the information transmission and communication services especially in areas with limited broadband access. 3G signal has advantages in mobility and availability and can compete with fixed media such as cable.

Given the advantages in overcoming geographic obstacles, ability and speed of data transmission, this research wants to add value to the 3G data service. Inverse multiplexing technique is a technique that is somewhat forgotten given the technical complexity in its implementation. Inverse Multiplexing or can also be referred to as a one-to-many, is a technique of splitting a high speed line into multiple lower speed lines[1]. This method will be used in this research as a method to obtain broadband capacity to transmit large data files from a server. Data packets will be passed through several channels to its direction.

2. Objectives and benefits

Referring to the need for broadband channels that are economic and flexible, the researchers propose the development of a system comprising a number of 3G channels to get the desired channel bandwidth. This broadband channel is specifically for connection to the Internet. This research provides alternative possibility to obtain a broadband transmission system that is economic and nomadic in accordance to the 3G services of the operator concerned.

The demands on recency and diversity of information that should be distributed and collected require a solution that can meet these demands. Information collected for distribution or further processing is generally multimedia which causes the need for broadband transmission. The location where the information is collected and resources are mobile. Conventional transmission such as cable, fiber optic and satellite in certain cases is not suitable. Mobility cannot be provided fiber optic cable. Satellite requires specialized equipment and high operational costs. This situation calls for alternative systems. 3G mobile communication system can connect to the Internet and has relatively high speed. Its operational cost is economic. The 3G service availability is quite spread out; hence, the need for mobility can be met through its creative use. The existence of a router that has a feature that allows the configuration of sending data packets simultaneously through a number of WAN channels enable transmission of data through a number of channels to the intended destination. Information will be sent on par with the use of high-speed transmission line. The benefits of this study provide the possibility of transmission of broadband by utilizing 3G services. This study supports the creative use of 3G technology.

3. Literature Review and Previous Works

3.1 3G Service

3G Service is the third generation mobile communications, a communication technology that evolved and developed to satisfy the demands for mobile broadband communications. The purpose of the introduction of 3G technology is to add the ability to achieve high data transfer speeds, improving the quality of service and support the needs of mobile Internet. ITU (International Telecommunication Union) defines 3G (Third Generation) as a technology that can be provide data transfer speed of 144 kbps for mobile user with speed of 100 km/h, has a data transfer speed of 384 kbps for users at walking pace and has a data transfer speed of 2 Mbps for stationary users. With data transfer speeds between 144kbps-2Mbps the 3G technology can serve broadband data services such as Internet, video on demand, music on demand, games on demand, and on the other multimedia demands such as video conferencing and video streaming.

Qiu, Zhu and Zhang[2] described the next evolution of 3G and 3.5G with its applications. 3G communication transmission speed is gradually developed specifically for the purpose of downloading information. Evolution to 4G implies that air interface will change. 3.5 G services provide increased capacity and spectral efficiency as found in WCDMA/ HSDPA (High Speed Download Packet Access). Operators that are GSM based will provide 3G high speed data services based on WCDMA and then adopted HSDPA to provide high speed broadband Internet access. Abdel El Al, Saadawi, and Lee[3] discussed the solution at the transport layer for heterogeneous aggregation through a

number of channels in mobile environment such as 3G services. For this purpose it is necessary to provide mechanisms that gives the mobile applications a logical channel that is approximately multiple of the nominal allotted bandwidth of each channel. In this case it is an extension to SCTP (Stream Control Transmission Protocol). SCTP uses multi-homed hosts for redundancy.

This research will use channels obtained from 3G data transmission services to the Internet. To obtain broadband services a number of 3G channels are used to implement inverse multiplexing concept. This system is used to upload or download large files to and from the Web Server.

3.2 Inverse Multiplexing

Inverse multiplexing is a technique that is opposite to the multiplexing. Multiplexing combines a number of low-speed transmission line to be channeled through a single high-speed channel. This technique enhances the usability of the low-speed channel so full capacity can be used. Inverse multiplexing technique using a transmission line having a low speed to distribute data from high-speed channel. Inverse multiplexing arises because of the need for broadband services through a telecommunications network that is basically made for the purposes of the demand for voice communication. This technique is referred to as inverse multiplexing. Duncanson[4] gives an overview of the inverse multiplexing for both high-speed packet switching technique known as Asynchronous Transfer Mode (ATM). As initially inverse multiplexing was a concept to achieve broadband for ATM (Asynchronous Transfer Mode ,for cell switching, the detail mechanism is different than IP based network and cellular network which is inherently not as stable as fixed wireline network. However, ATM services are not growing as expected and Inverse Multiplexing concept becomes obscure. Hence, the proposed solution is considered novel as it is applied in a TCP/IP environment

Fredette[5] gave an overview of the inverse multiplexing where the concept of inverse multiplexing is used by a connection oriented service for data packet. To get a broadband connection is need for special protocols that can fulfill a number of technical parameters. The principle used is sending data packets through a separate channel. In the package recipient of a number of these channels will be assembled into a packet flow as before. Issues that must be considered were reassembly and time delay. The BONDING (Bandwidth On Demand Interoperability Group) standard has not been able to meet the need for an appropriate solution.

Inverse multiplexing is one concept to address the need for high-speed services that is is quite interesting considering that high-speed transmission channel is always available or too expensive. There are advantages in the use of inverse muxing techniques to get high speed broadband services. The demand for high speed broadband increased in view of the applications and services that require a large size and high-speed transmission. Interface for connection Local Area Networking relatively inexpensive and high-speed broadband and is generally 100 Mbps ie speed Ethernet. Newer Ethernet technology has a speed of more than one Gigabit per second. High speed applications that require large bandwidth is a common demand. Inverse multiplexers can provide national and international connections for high-speed broadband connection to a variety of applications such as video conferencing for high quality, and high-speed back-up with economic costs.

Other researchers are Funk et al[6] conducted a study using multiplexing techniques for short-range wireless connections such as via infrared and blue tooth. Their research shows the benefits along with a number of further research needs to be performed. The research Snoeren[1] deals with the inverse multiplexing for Wide Area Wireless Network.

The characteristics of wireless channels are different compared to fixed lines such as cable or optical fibers concerning to its capacity and time delay (latency). Fixed line characteristics are basically stable whereas wireless channels are dynamic. One of the aspects that was studied is the line balancing quality to perform load balancing. TCP or UDP transports data. Limitations of bandwidth of wide-area wireless access networks (WWANs) often interfere with services such as streaming video audio or high-resolution images applications. Inverse multiplexing can provide broadband connectivity. It separates the data packets to a number of physical connections and establish a logical broadband connection. Most WWAN technology shares channels that have non uniform characteristics especially the available bandwidth, latency, and loss rates. Snoeren[1] proposed a solution that uses the adaptive inverse multiplexing based on balancing the link quality. The principle is to use performance measures for

scheduling and selection of channels used at any time. Similarly Brassil[7,8] discussed the use of a number of communication channels into a high-capacity WAN connection. The discussion is primarily concerned with consequence of using heterogeneous such as ADSL, cable modem line, fiber optics, and so forth including protocols[9,10,11]

MikroTik is a manufacturer that produces economic routers that have advanced features. MikroTik Router provides convenience to the user to perform monitoring or packet tracking for troubleshooting. It can be remotely controlled with the help of Winbox as the program. Winbox is also a Graphic User Interface to configure the router. The router has the ability to perform load balancing load. Load balancing is a technique to divide the load to multiple connections or links that exist to obtain the necessary throughput. This technique will measure the load on a channel, if the channel is congested, packets will be routed to another track to accelerate data transmission or keep the throughput stable.

MikroTik has several techniques to perform load balancing such as PCC and NTH. PCC (Per Connection Classifier) divide a data stream to a balanced path complying to the specifications. NTH is based on round robin, but the method is improved as the load is divided sequentially. Packets flow from one gateway to another gateway. NTH define persistent user session as a user who uses the IP address in its communication. To run the load balancing process MikroTik performs a technique called Mangle. Mangle is used to mark a packet. The packet is directed in accordance with the process set. Flagging will help Micro Tik router to perform a variety of specialized tasks within the router. Mangle is not sent out but is only used by the router in question. Sometimes mangle is also used to change the IP header such as TOS or TTL. Micro Tik has a number in the form of CLI commands that can be used to execute a particular process or provide certain statistical data.

4. Methodology

4.1 System Diagram

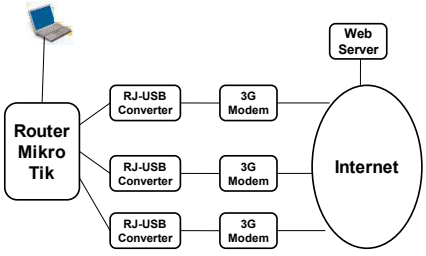


Fig. 1 System Diagram

Data communication system using 3G must communicate to a server in the Internet. The destination of the file must be the server. A receiver can then download the file of interest from this server. Currently, 3G does not provide peer to peer communication. End User devices (User Devices such as laptop or other computer to a server connected to the Internet. The connection must be through a router that to the Internet. The Router will use 3G service to reach the server. The current research uses MikroTik router. MikroTik has ports that can be used as WAN. These ports physically are RJ-45. Consequently, to connect to a 3G modem a converter to USB is needed. A 3G modem has USB physical connector. To perform inverse multiplexing several 3G lines are used as transmission media. Inverse multiplexing is performed in the MikroTik router that is configured to accordingly. The configuration commands are supplied via scripts. Packets are managed by MikroTik according to the configuration set. Packets are delivered to the server through the modems to the Internet. Throughput will increase as if the router has a higher speed line. The router will balance the load according to the configuration defined via the scripts. The role of MikroTik router in this research is performing inverse multiplexing to the 3G lines. Configuration of MikroTik is set through Command Line Interface (CLI) via MikroTik Terminal. Winbox is software to communicate to MikroTik remotely.

MikroTik script is used to configure the router following a predefined scheme. The first script cleans the router from existing configurations and sets a new configuration to perform inverse multiplexing. NTH load balancing is

the algorithm that can send packets to the outgoing lines evenly or balanced. The system will behave similar to an inverse multiplexer. Three ports are used for WAN. To ensure that sending of packets is evenly distributed, MikroTik will make each packet. The process is called mangle. A script is loaded to define how the mangle should work.

4.2 Measurements

4.2.1 Measurement of the effect of bandwidth to transfer time

To measure the effect of bandwidth to transfer time, bandwidth of the line will be limited. The time to transfer a file of selected bandwidth will be measured. The measurement is repeated several times and the average value can be noted. The same file is transferred via the selected line, but no bandwidth limitation. The bandwidth limitation is applied to the USB to RJ-45 converter. The converter has bandwidth control function. Bandwidth Control is enabled to choose bandwidth size. The data file for this experiment was just an ordinary file that was reasonably large. Nothing special of the file itself. The size was large to show the effect of using multiple lines to increase the transmission lines to obtain bandwidth.

4.2.2 Measurement of the effect of number of lines to transfer time

To measure the effect of number lines to transfer time, files of selected size is send to the router that has two and three 3G lines. The time it took to arrive at the destination will be noted. The process is repeated to arrive to an average. The selected file size is 10 MB and 100 MB. The transfer time represent the band width. The smaller the transfer time the higher the bandwidth. If the load balancing of the router performs according to its design the bandwidth will be multiple of 3G capacity and transfer time could show the result.

The measurement could proof that the inverse multiplexing concept can be implemented using the combination of Router Micro Tik and TP-Link. It can also show that applying inverse multiplexing a novel broadband solution can be provided to cover the need for high capacity transmission in a short time. It is not intended for permanent broadband link due to the instability and dynamic bandwidth.

5. Results and Discussions

5.1 Data transfer time with multiple lines

These measurements are meant to indicate whether the system developed can serve as an inverse multiplexer that uses the N channel capacity to get a virtual channel with ideal capacity N X M bit per second. Each channel capacity is M bit per second. Measurements for inverse multiplexer of two channels and three channels were made. The transfer time of multiple lines for dowloading a 10MB file was compared to a single line transfer time. The time is measured semi manually. It was taken from the print-out of the measured data. The following results were tabulated as Table-01. One 3G line needed 87 seconds. The result showed that the two modems simultaneously transfer data packets. The traffic at the two lines is balanced. It took in average 32 seconds to transfer a10MB file. The test was repeated work for three lines. The average transfer time to transfer a 10MB file via 3 lines is 12 seconds.

5.2 Data transfer time of a large file with multiple lines.

The next measurement concern the transfer of large file, in this experiment 100MB, is then performed to determine the performance of the system using one, two or three 3G channels. Transfer time comparison of multiple lines is tabulated in Table-02. The smaller the bandwidth the longer it takes to transmit takes to transmit the data. It looks sending data by using three modems is faster than the one or two modems. The most significant difference lies in the use of three modems.

6. Conclusions and Future Works

The results of this research indicated that inverse multiplexing creates a broadband line to the Internet through multiple 3G service. 3G service by itself has bandwidth limitations that prevent the quick transfer of large files. Sample no. 1, 2, 3 etc., were the same file being sent as samples to find the effect of multiple lines consisting of 3G lines. Using multiple 3G lines increases the bandwidth but it will not be a linear multiple of the line capacity. These limitations can be overcome by adding the number of 3G lines used to transmit data. Data packets can be simultaneously transmitted via several paths from a server on the Internet. This research capitalizes Micro Tik router and TP-Link capabilities to jointly perform the function of inverse multiplexing. The research priority is to prove that the concept of using a number of channels with 3G services together will generate broadband services. Given that the channel capacity of 3G unstable, with the understanding that the speed it offers can dynamically change either because of the nature of electromagnetic wave itself but also the traffic within the Internet itself. The results did show an increase in channel capacity although not exactly linear. 3G service for broadband connection to the Internet can be justified, due to its convenience and price competitive.

Table-01 Transfer time a 10MB file

Sample Number	Transfer Time of 10 MB File in seconds		
	Single 3G Line	Two 3G Lines	Three 3G Lines
1	114	30	12
2	97	45	10
3	121	32	12
4	98	30	13
5	112	26	13
6	64	25	12
7	62	22	11
8	85	38	11
9	58	46	12
10	59	30	12

Table 02 Transfer time a 100 MB

Sample Number	Transfer Time of 100 MB File in seconds		
	Single 3G Line	Two 3G Lines	Three 3G Lines
1	2547	1779	1665
2	3216	1802	783
3	3652	1799	1122
4	3198	2175	777
5	3326	2051	939
6	2447	2057	1066
7	2949	2268	797
8	2686	2002	1022
9	3796	1829	1093
10	3274	1926	971

It is recommended to perform further research in the inverse multiplexing concept to understand the most efficient bundling of 3G lines in terms of capacity, reliability and service quality. It is also of interest to find if the 3G line should be subscribed from one operator or multiple operators. It need further research to find alternative inverse multiplexing technique using different routers or devices. This study focuses for downloading large files from a web server on the Internet, but it could be developed further for video streaming.

References

1. Snoeren, A.C. 1999. Adaptive inverse multiplexing for wide-area wireless networks. Global Telecommunications Conference, 1999. GLOBECOM '99 , vol.3, pp.1665-1672
2. Qiu, R.C., Zhu, W., Zhang, Y. Third-Generation and Beyond (3.5G) Wireless Networks and Its Applications. IEEE International Symposium.Circuits and Systems, 2002. ISCAS 2002, vol.1, pp-41-44
3. Ahmed Abd El Al; Saadawi, T.; Lee, M. 2004. A transport layer load sharing mechanism for mobile wireless hosts,. Proceedings of the Second IEEE Annual Conference on Pervasive Computing and Communications Workshops, 2004, pp 87 - 91
4. Duncanson, J. 1994. Inverse multiplexing. IEEE Communications Magazine, IEEE , vol.32, no.4, pp.34,41
5. Fredette, P.H. 1994. The past, present, and future of inverse multiplexing, 1994. IEEE Communications Magazine, vol.32, no.4, pp.42,46
6. Funk, J.C.; Duffin, H.R.; Dai, L.; Knutson, C.D. 2003. Inverse multiplexing in short-range multi-transport wireless communications. IEEE Wireless Communications and Networking, vol.2, no., pp.757,762 vol.2, 20-20 March 2003
7. Brassil, J. 2005. An empirical study of bonded heterogeneous WAN access links. The 14th IEEE Workshop on Local and Metropolitan Area Networks, 2005.
8. Brassil, J.; et al. 2008. Improving VPN performance over multiple access links. 5th International Conference on Broadband Communications, Networks and Systems, 2008, pp. 649 - 656
9. Kyu-Han Kim; Shin, K.G. 2007. PRISM: Improving the Performance of Inverse-Multiplexed TCP in Wireless Networks. IEEE Transactions on Mobile Computing, Volume: 6, Issue: 12, pp 1297 - 1312
10. Kokku, R.; Bohra, A.; Ganguly, S.; Venkataramani, A. 2007. A Multipath Background Network Architecture. INFOCOM 2007. 26th IEEE International Conference on Computer Communications. pp. 1352 - 1360
11. Lane, J.R.; Nakao, A. 2008. Best-Effort Network Layer Packet Reordering in Support of Multipath Overlay Packet Dispersion. IEEE Global Telecommunications Conference, 2008. pp. 1 - 6